MATH 1400 · QUIZ 5 · SIDE 1 — Arc Length · Surface Area · Average Value · Centroids / Center of Mass

ARC LENGTH

$$L = \int_a^b \sqrt{1 + (y')^2}\, dx \quad \text{or} \quad \int_c^d \sqrt{1 + (x')^2}\, dy \quad \text{(x=g(y))}$$

Parametric: $L = \int_\alpha^\beta \sqrt{(x')^2 + (y')^2}\, dt$

Goal: make the radicand a perfect square. Steps: ① find y' ② square ③ add 1 ④ factor ⑤ √ cancels the square ⑥ integrate.

- **Trick A ($A - B$ form):** if $y' = A - B$ and $AB = \frac{1}{4}$, then $1 + y'^2 = (A + B)^2$. Flip the middle sign. e.g. $y' = x^3/4 - x^{-3} \Rightarrow \sqrt{} = x^3/4 + x^{-3}$.
- Typical curves: $\frac{2}{3}x^{3/2} - \frac{1}{2}x^{1/2}$ (y' = $x^{1/2} - \frac{1}{4}x^{-1/2}$), $\frac{x^3}{6} + \frac{1}{2x}$ ($\sqrt{} = \frac{x^2}{2} + \frac{1}{2x^2}$), $x^2 - \frac{\ln x}{8}$, $\frac{x^4}{16} + \frac{1}{2x^2}$.
- **Trick B (one term):** $y = c(\dots)^{3/2} \Rightarrow y'$ has a √, so y'^2 is polynomial. e.g. $y = \frac{4}{3}x^{3/2} \Rightarrow \sqrt{1 + 4x}$; $y = \frac{2}{3}(4x^2 + \frac{1}{2})^{3/2} \Rightarrow y' = 8x\sqrt{4x^2 + \frac{1}{2}}$, $1 + y'^2 = 256x^4 + 32x^2 + 1 = (16x^2 + 1)^2 \Rightarrow L = \int_0^3 (16x^2 + 1)dx = 147$.
- **ln curves:** $y = \ln\cos x \Rightarrow ds = \sec x\, dx$, $\int \sec = \ln|\sec x + \tan x|$. $y = \ln(1 - x^2) \Rightarrow y' = \frac{-2x}{1-x^2}$, $ds = \frac{1+x^2}{1-x^2}dx = \left(-1 + \frac{2}{1-x^2}\right)dx \to -x + \ln\frac{1+x}{1-x}$.
- **Parametric trick:** $x'^2 + y'^2$ often a square. $x = \frac{1}{3}(2t-1)^{3/2}, y = \frac{1}{2}(t-1)^2$: $x'^2 + y'^2 = (2t-1) + (t-1)^2 = t^2 \to L = \int_1^3 t\, dt = 4$.
- If x=g(y) is easier (or y' blows up), integrate in **dy**. Check limits are in the right variable!

SURFACE AREA OF REVOLUTION

$$SA = \int 2\pi\,(\text{radius})\, ds, \quad ds = \sqrt{1 + y'^2}\, dx = \sqrt{1 + x'^2}\, dy = \sqrt{x_t^2 + y_t^2}\, dt$$

x-axis ⇒ radius = y | y-axis ⇒ radius = x

Radius must be written in the same variable as the integration variable (x=g(y), or y=f(x), or in t). **Axis → radius; given form → variable/ds.**

- **Cancellation trick:** radius×ds often simplifies. $y = \sqrt{x}$: $y\, ds = \sqrt{x} \cdot \frac{\sqrt{4x+1}}{2\sqrt{x}} = \frac{1}{2}\sqrt{4x+1} \to$ u=4x+1. SA(0..2)=$\frac{13\pi}{3}$.
- **Circle trick:** $y = \sqrt{R^2 - x^2}$ about x-axis ⇒ $y\, ds = R\, dx$ so $SA = 2\pi R\,(\text{width})$. e.g. R=5, x∈[3,4] → 10π. (Same as zone of sphere $2\pi Rh$.) $x = \sqrt{4 - y^2}$, y-axis, [−1,1] → $2\pi \cdot 2 \cdot 2 = 8\pi$.
- **Perfect-square ds:** $x = \frac{1}{3}(y^2 + 2)^{3/2}$ (x-axis): $x' = y\sqrt{y^2 + 2}$, $1 + x'^2 = (y^2 + 1)^2 \Rightarrow ds = (y^2 + 1)dy$, $SA = 2\pi\int y(y^2 + 1)dy$ = $\frac{21\pi}{2}$ on [1,2].
- **u-sub:** $\int x\sqrt{1 + 4x^2}\, dx$: $u = 1 + 4x^2$. $x = y^3$ (y-axis): $2\pi\int y^3\sqrt{1 + 9y^4}dy$, $u = 1 + 9y^4$. $y = 1 - x^2$ (y-axis): $2\pi\int x\sqrt{1 + 4x^2}\, dx$.
- $y = \cosh x = \frac{e^x + e^{-x}}{2}$: $ds = \cosh x\, dx$, $y\, ds = \cosh^2 x = \frac{1 + \cosh 2x}{2}$.
- **Parametric:** x-axis: $\int 2\pi\, y(t)\sqrt{x_t^2 + y_t^2}\, dt$; y-axis: $\int 2\pi\, x(t)\sqrt{\dots}\, dt$. Ex: $x = 2\sqrt{t}, y = 1 - t$ (y-axis): $ds = \frac{\sqrt{1+t}}{\sqrt{t}}dt \Rightarrow 4\pi\int\sqrt{1 + t}$ =168π on [0,15].
- **Sanity:** arc length & SA are positive; SA should not have a sign issue—if $y < 0$ use |radius|.

AVERAGE VALUE

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\, dx.$$ MVT for integrals: $\exists c \in (a,b)$: $f(c) = f_{avg}$ (solve $f(c) = f_{avg}$, keep root inside [a,b]).

Parametric: y_{avg} w.r.t. x = $\frac{1}{x(\beta) - x(\alpha)}\int_\alpha^\beta y\, x'(t)\, dt$ (note $dx = x'dt$; $x(\beta) - x(\alpha)$ can be negative).

- $\frac{1}{x^2-1} = \frac{1}{2}\left[\frac{1}{x-1} - \frac{1}{x+1}\right]$: $\frac{1}{3}\int_2^5 \frac{6}{x^2-1} = \ln 2$.
- $\frac{16}{(x^2+4)^{3/2}}$ on $[0, 2\sqrt{3}]$ → trig sub, $f_{avg} = 1$.
- Half-angle: $\sin^2 = \frac{1-\cos 2t}{2}$, $\cos^2 = \frac{1+\cos 2t}{2}$. Semicircle $x = \cos t, y = \sin t$: $y_{avg} = \pi/4$.

CENTROIDS & CENTER OF MASS

Region between top f and bottom g, $a \le x \le b$, density $\rho(x)$ (ρ=1 ⇒ centroid):

$m = \int \rho\,(f - g)\, dx \quad M_y = \int x\,\rho\,(f - g)\, dx \quad M_x = \frac{1}{2}\int \rho\,(f^2 - g^2)\, dx$

$\bar{x} = \frac{M_y}{m} \quad \bar{y} = \frac{M_x}{m}$ ($M_y \to \bar{\mathbf{x}}$, $M_x \to \bar{\mathbf{y}}$)

Rod: $m = \int \rho\, dx$, $\bar{x} = \frac{\int x\rho\, dx}{\int \rho\, dx}$.

Discrete: $\bar{x} = \frac{\sum m_i x_i}{\sum m_i}$.

- **Density depends on x only** (vertical strips). Density pulls center toward heavy side (ȳ uses ρf^2, not just ρf).
- **Symmetry first:** symmetric about y-axis ⇒ x̄=0; about x-axis ⇒ ȳ=0; sym. about y=x ⇒ x̄=ȳ. Even region about y-axis ⇒ integrate 2× right half. (ρ=x breaks symmetry!)
- Find intersections to get a,b. Top = larger f on interval (cos vs sin on $[\frac{\pi}{4}, \frac{3\pi}{4}]$: sin on top; $\sin^2 - \cos^2 = -\cos 2x$).
- **Standard centroids** (use instead of integrating):

Rectangle $w \times h$	$(w/2, h/2)$
Semicircle (upper, r)	$(0, \frac{4r}{3\pi})$
Quarter circle (1st quad)	$(\frac{4r}{3\pi}, \frac{4r}{3\pi})$
Right Δ (0,0),(b,0),(0,h)	$(b/3, h/3)$
Any triangle	avg of 3 vertices
Areas	disk πr^2, semi $\frac{1}{2}\pi r^2$, quarter $\frac{1}{4}\pi r^2$, Δ $\frac{1}{2}bh$

Composite: $\bar{x} = \frac{\sum A_i\bar{x}_i}{\sum A_i}$, $\bar{y} = \frac{\sum A_i\bar{y}_i}{\sum A_i}$. Holes = **negative area**.

Ex: tri (−2,0),(0,0),(0,2) [A=2, (−⅔,⅔)] + quarter circle r=2 [A=π, (8/3π, 8/3π)] ⇒ $\bar{x} = \frac{4}{3(\pi+2)}$, $\bar{y} = \frac{4}{\pi+2}$.

Answers to check: parabola $y = 4 - x^2$/x-axis: $(0, \frac{8}{5})$; $y = x$ & x^2: $(\frac{1}{2}, \frac{2}{5})$, with ρ=x: $(\frac{3}{5}, \frac{1}{2})$; $\sqrt{x}, x^2$, ρ=35x: x̄=4/7; $2x^2$ & $3 - x^2$: $(0, \frac{8}{5})$; quarter disk r=2, ρ=x: ȳ=¾; triangle (0,0),(0,4),(6,0): x̄=2; $\sqrt{2}/4$ for cos/sin.

MATH 1400 · QUIZ 5 · SIDE 2 — Probability & Density Functions · Integration Toolkit

PDF BASICS

Valid PDF: ① $f(x) \geq 0$ ② $\int_{-\infty}^{\infty} f(x)\,dx = 1$ (find unknown constant C by setting this =1).

$P(a \leq X \leq b) = \int_a^b f\,dx$. $P(X = a) = 0$ so < vs ≤ doesn't matter. $P(X > a) = 1 - P(X \leq a)$. f can exceed 1 (it's a density, not a probability).

Mean $\mu = \int x f(x)\,dx$ **Median** m: $\int_{-\infty}^{m} f = \frac{1}{2}$ (use the easier end: $\int_m^{\text{right}} f = \frac{1}{2}$)
Variance $\sigma^2 = \int x^2 f\,dx - \mu^2$ (easier than $\int (x-\mu)^2 f$) $\sigma = \sqrt{\sigma^2}$
Percentile x_p: $F(x_p) = p$. All integrals only over the <u>support</u> (where f≠0).

- **Median recipe:** integrate f from left edge to m, set = ½, solve for m. e.g. $g = 3\sin 3x$ on $[0, \frac{\pi}{6}]$: $1 - \cos 3m = \frac{1}{2} \Rightarrow m = \frac{\pi}{9}$. $f = \frac{x^2}{39}$ on [2,5]: $m = \sqrt[3]{133/2}$.
- **Mean/σ recipes:** $\frac{3}{x^4}$ on $[1,\infty)$: $\mu = \frac{3}{2}$, $E[X^2] = 3$, $\sigma^2 = 3 - \frac{9}{4} = \frac{3}{4}$, $\sigma = \frac{\sqrt{3}}{2}$. $\frac{2\ln x}{x}$ on $[1,e]$: $\mu = \int 2\ln x\,dx = 2$. $3\sin 3x$: $\mu = \frac{1}{3}$.
- Improper: $\int_1^{\infty} x^{-p}dx = \frac{1}{p-1}$ $(p > 1)$. Write as $\lim_{b\to\infty}$ and say so.

SPECIAL DENSITIES

Exponential (x≥0): $f = ke^{-kx} = \frac{1}{\mu}e^{-x/\mu}$, $\mu = \frac{1}{k}$, $\sigma = \mu$
CDF $F = 1 - e^{-x/\mu}$; $P(X > x) = e^{-x/\mu}$; **median** $= \mu\ln 2$ (e.g. $f = \frac{1}{3}e^{-x/3}$: median $3\ln 2$). Bank wait μ=8: $P(X > 10) = e^{-5/4}$, med $8\ln 2$.
Normal $f = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$; standard normal $\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ (μ=0,σ=1, symmetric: mean=median=μ). $z = \frac{x-\mu}{\sigma}$.
Uniform on [a,b]: $f = \frac{1}{b-a}$, $P(c \leq X \leq d) = \frac{d-c}{b-a}$.

$Cxe^{-x/2}$, **x≥0:** $\int_0^{\infty} xe^{-x/2}dx = 4 \Rightarrow C = \frac{1}{4}$.

CDF

$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$ (use dummy t). $f = F'$
Piecewise: $F = 0$ (left of support), $\int_a^x f(t)dt$ (on support), 1 (right).
Check $F(a) = 0, F(b) = 1$.
$P(X \leq b) = F(b)$, $P(X > a) = 1 - F(a)$, $P(a \leq X \leq b) = F(b) - F(a)$. Median: $F(m) = \frac{1}{2}$.
Properties: 0≤F≤1, non-decreasing, limits 0 and 1, continuous.

- Given F → differentiate for f (product rule!). $F = 1 - (1 + x)e^{-x} \Rightarrow f = xe^{-x}$; then $P(X > 3) = 4e^{-3}$.
- $f = \frac{2}{27}x(x-1)$ on [1,4]: $F = \frac{2x^3 - 3x^2 + 1}{81}$; $P(2 \leq X \leq 3) = \frac{23}{81}$.

GEOMETRIC PROBABILITY

$P(A) = \frac{\text{Area}(A)}{\text{Area}(R)}$ (uniform on R).
Complement: $1 - P$. π cancels for circles—use r^2 ratios. Dart (R=12): bullseye r=2: $\frac{1}{36}$; ring 6–8: $\frac{28\pi}{144\pi} = \frac{7}{36}$; outside r=8: $1 - \frac{4}{9} = \frac{5}{9}$.
Meeting problem: square side T, wait w: $P = 1 - \frac{(T-w)^2}{T^2}$ (T=60,w=15 → $\frac{7}{16}$). Region $|x - y| \leq w$.
Under curve in rectangle: $\frac{\int(\text{curve})}{\text{rect area}}$.
Distance-from-center R: $F(r) = \frac{r^2}{144}$, $f = \frac{r}{72}$, mean 8, median $6\sqrt{2}$.

INTEGRATION TOOLKIT (shows up everywhere)

Trig sub ★ for $(x^2 + a^2)^{3/2}$ **/** $(a^2 - x^2)^{3/2}$ **(memorize results):**
$\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{x}{a^2\sqrt{x^2+a^2}}$
$\int \frac{dx}{(a^2-x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2-x^2}}$
(derive: $x = a\tan\theta$ or $a\sin\theta$). Ex: $f = \frac{75/4}{(25-x^2)^{3/2}}$, $P(X \leq \frac{5}{2}) = \frac{3}{4}$. $\frac{5/2}{\sqrt{25-25/4}} = \frac{\sqrt{3}}{4}$ (a=3,b=4).
u-sub: $\int x(\dots)^{1/2}$ with $u =$ inside; $\int u^{1/2}du = \frac{2}{3}u^{3/2}$. **Change limits** with u!
Parts: $\int u\,dv = uv - \int v\,du$ (LIATE). $\int xe^{-kx}dx = -\frac{x}{k}e^{-kx} - \frac{1}{k^2}e^{-kx}$; $\int_0^{\infty} x^n e^{-kx}dx = \frac{n!}{k^{n+1}}$.
$\int \ln x\,dx = x\ln x - x$. $\int x\sin 3x$: u=x.
Partial fractions: $\frac{1}{x^2-a^2} = \frac{1}{2a}\left[\frac{1}{x-a} - \frac{1}{x+a}\right]$.
$\int \sec x = \ln|\sec x + \tan x|$,
$\int \sin ax = -\frac{\cos ax}{a}$, $\int \frac{dx}{x} = \ln|x|$,
$\int e^{kx} = \frac{e^{kx}}{k}$. $\sin^2, \cos^2$ half-angle.

QUICK CHECKLIST / TRAPS

- Arc/SA: **first look for the perfect square**; if not square, check you differentiated right.
- SA: axis decides radius; **don't** leave y in a dx-integral (substitute f(x)). 2π out front.
- Centroid: use symmetry & standard shapes; M_x has ½ and squares; divide by area (or mass).
- Density → center of mass shifts toward larger ρ. Sanity check that $(\bar{x}, \bar{y})$ lies inside region.
- Probability: check ∫f=1 first when asked for constant. Mean uses xf; variance uses $x^2 f - \mu^2$.
- Exact answers, no calculator: leave π, $\ln 2$, radicals, e^{-k}.

Known answers (check your setups): L: $\frac{4}{3}x^{3/2}$[0,2]=$\frac{13}{3}$; $\frac{x^3}{6} + \frac{1}{2x}$[1,3]=$\frac{14}{3}$; $\ln(1-x^2)$[0,½]=$\ln 3 - \frac{1}{2}$; $\ln\cos x$[0,π/3]=$\ln(2+\sqrt{3})$; $\frac{2}{3}x^{3/2} - \frac{1}{2}x^{1/2}$[¼,4]=6; $\frac{x^4}{16} + \frac{1}{2x^2}$[1,2]=$\frac{21}{16}$.
SA: $\frac{x^3}{3}$(x-ax, [0,$\sqrt[4]{8}$])=$\frac{26\pi}{9}$; $x = \frac{2}{3}y^{3/2}$(x-ax,[0,3])=$\frac{232\pi}{15}$ (u=1+y, **y=u−1**); cosh[0,ln2]=$\frac{15\pi}{16} + \pi\ln 2$; $1 - x^2$(y-ax)=$\frac{7\pi}{6}$; y^3(y-ax)=$\frac{7\pi}{27}$; $\sqrt{x}$[0,2]=$\frac{13\pi}{3}$.